

Surface Mount Frequency Mixer

Level 7 (LO Power +7 dBm) 20 to 1000 MHz

JMS-2+

Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power	50mW
IF Current	40mA
Permanent damage may occur if any of these limits are exceeded.	

Pin Connections

LO	6
RF	3
IF	2
GROUND	1,4,5

Features

- excellent L-R isolation, up to 63 dB typ.
- miniature surface mount
- J-leads for strain relief and excellent solderability

Applications

- up & down converters for receivers & transmitters
- VHF/UHF
- cellular/ISM/GSM



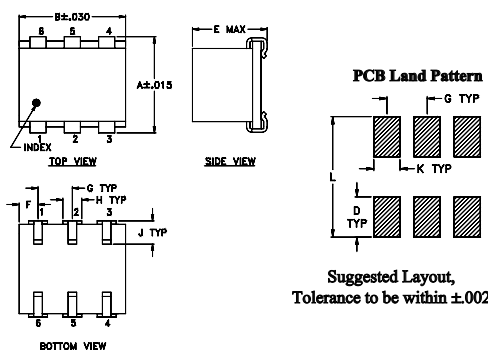
Generic photo used for illustration purposes only

CASE STYLE: BH292

+RoHS Compliant

The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications

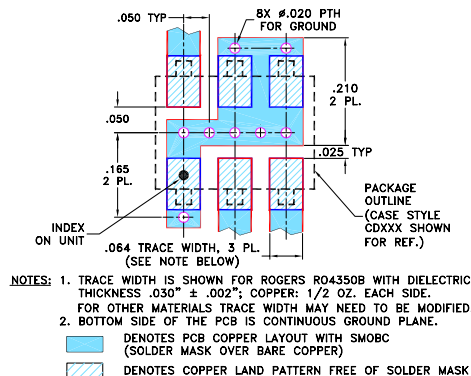
Outline Drawing



Outline Dimensions (inch)

A	B	C	D	E	F	G
.280	.310	--	.100	.225	.055	.100
7.11	7.87	--	2.54	5.72	1.40	2.54
H	J	K	L	wt		
.047	.065	.065	.300	grams		
1.19	1.65	1.65	7.62	0.45		

Demo Board MCL P/N: TB-03 Suggested PCB Layout (PL-052)



Notes

- Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
- Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
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Electrical Specifications

FREQUENCY (MHz)		CONVERSION LOSS (dB)				LO-RF ISOLATION (dB)						LO-IF ISOLATION (dB)						IP3 at center band (dBm)	
LO/RF	IF	Mid-Band		Total Range Max.		L	M	U				L	M	U					
		$\overline{X}$	σ																Max.
		f_L - f_U																	
20-1000	DC-1000	7.0	.15	8.4	9.5	63	40	50	28	35	20	56	30	47	22	37	20	17	

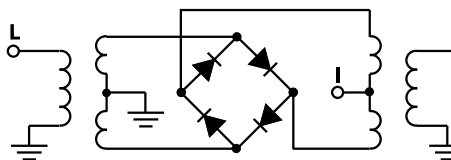
1 dB COMP: +1 dBm typ.
Phase detection, positive polarity

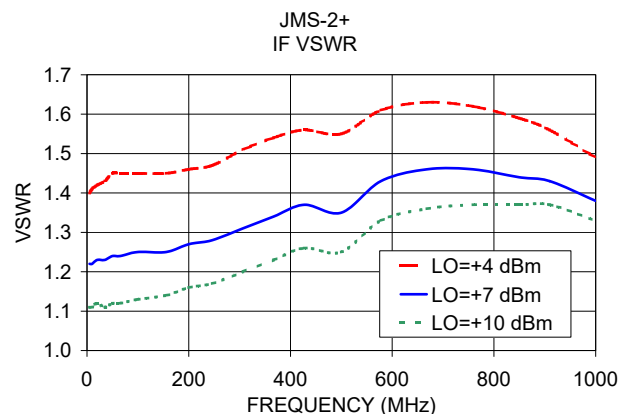
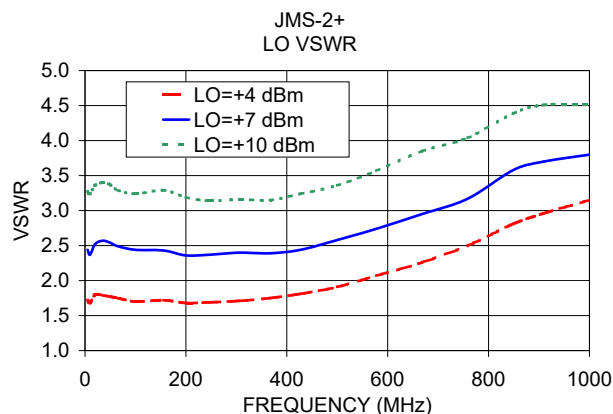
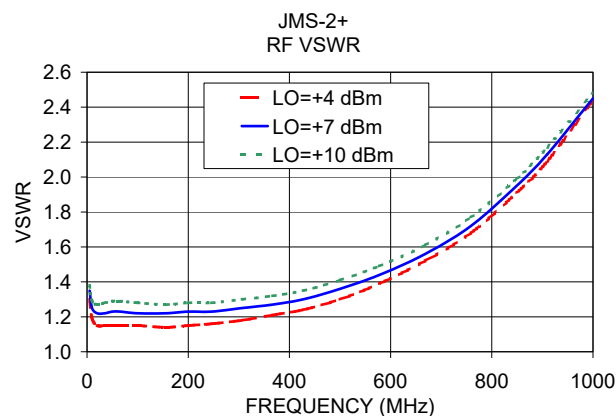
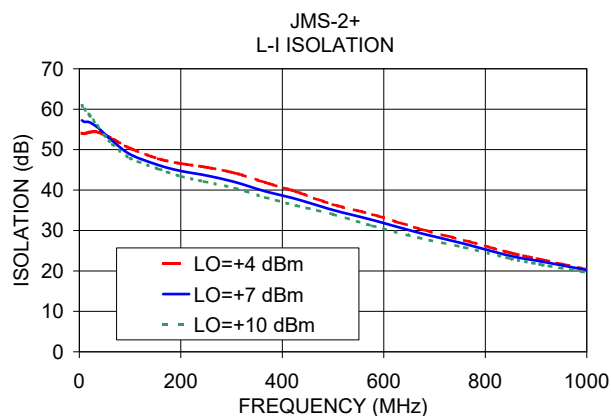
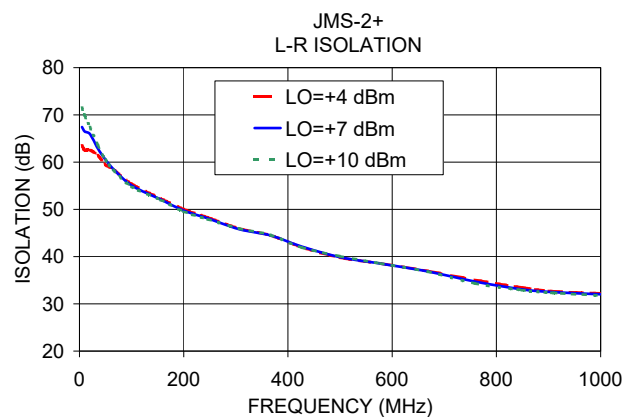
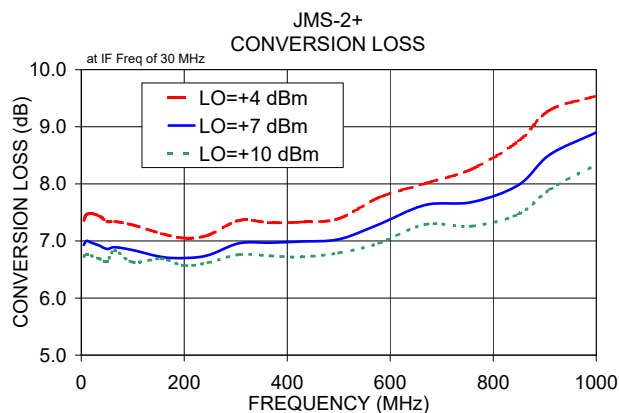
L = low range [f_L to 10 f_L]
m = mid band [$2f_L$ to $f_U/2$]
M = mid range [10 f_L to $f_U/2$]
U = upper range [$f_U/2$ to f_U]

Typical Performance Data

Frequency (MHz)		Conversion Loss (dB)	Isolation L-R (dB)	Isolation L-I (dB)	VSWR RF Port (:1)	VSWR LO Port (:1)
RF	LO	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm
5.00	35.00	6.93	67.42	57.23	1.35	2.44
10.00	40.00	7.00	66.49	56.87	1.25	2.37
20.00	50.00	6.97	65.94	56.78	1.22	2.52
35.15	65.15	6.92	62.92	55.59	1.22	2.57
50.00	80.00	6.86	60.51	53.87	1.23	2.54
65.30	95.30	6.89	58.52	52.15	1.23	2.49
100.00	70.00	6.84	55.07	48.85	1.22	2.44
155.76	125.76	6.72	52.10	46.19	1.22	2.43
200.00	170.00	6.70	49.75	44.71	1.23	2.36
246.21	216.21	6.75	48.10	43.71	1.23	2.37
306.52	276.52	6.96	45.81	41.99	1.25	2.40
366.82	336.82	6.97	44.49	39.66	1.27	2.39
427.12	397.12	6.99	42.09	37.81	1.30	2.44
500.00	470.00	7.03	39.89	35.06	1.36	2.58
577.88	547.88	7.29	38.51	32.59	1.44	2.74
668.33	638.33	7.63	36.87	29.46	1.56	2.95
758.79	728.79	7.68	34.71	26.72	1.72	3.17
849.24	819.24	7.98	33.07	23.71	1.95	3.58
909.54	879.54	8.51	32.47	22.35	2.13	3.70
1000.00	970.00	8.90	32.02	20.21	2.45	3.80

Electrical Schematic





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Frequency Mixer

JMS-2+

Typical Performance Data

RF (IN) (MHz)	LO (MHz)	CONVERSION LOSS IF FIXED @IF(OUT)=30MHz (dB)		
		@LO (dBm)		
		+4	+7	+10
10.1	40.1	7.00	6.08	5.90
49.8	79.8	7.45	6.98	6.79
89.5	119.5	7.52	7.06	6.84
129.2	159.2	7.46	7.03	6.80
168.9	198.9	7.44	7.07	6.83
208.6	238.6	7.40	7.02	6.79
248.3	278.3	7.43	6.96	6.81
287.9	317.9	7.34	6.97	6.79
327.6	357.6	7.38	7.00	6.83
367.3	397.3	7.34	7.01	6.83
407.0	437.0	7.30	7.05	6.81
446.7	476.7	7.32	7.05	6.80
486.4	516.4	7.38	7.04	6.85
526.1	556.1	7.37	7.07	6.78
565.8	595.8	7.43	7.05	6.81
585.6	615.6	7.41	7.04	6.81
625.3	655.3	7.46	7.06	6.85
645.2	675.2	7.44	7.14	6.91
684.9	714.9	7.47	7.15	6.89
704.7	734.7	7.47	7.13	6.88
744.4	774.4	7.58	7.19	6.95
764.3	794.3	7.56	7.20	7.03
803.9	833.9	7.65	7.26	7.09
823.8	853.8	7.72	7.31	7.08
863.5	893.5	7.83	7.44	7.25
883.3	913.3	7.87	7.45	7.21
923.0	953.0	7.97	7.54	7.36
942.9	972.9	8.02	7.61	7.34
982.6	1012.6	8.13	7.67	7.36
1002.4	1032.4	8.21	7.70	7.42
1042.1	1072.1	8.30	7.81	7.54
1061.9	1091.9	8.48	7.88	7.55
1101.6	1131.6	8.55	8.00	7.65
1121.5	1151.5	8.56	8.04	7.68
1161.2	1191.2	8.69	8.14	7.80
1181.0	1211.0	8.83	8.26	7.89
1220.7	1250.7	8.83	8.23	7.90
1240.6	1270.6	8.99	8.43	8.07
1280.3	1310.3	9.08	8.54	8.12
1300.1	1330.1	9.15	8.61	8.26

RF (IN) (MHz)	LO (MHz)	IP3 INPUT (dBm)		
		@LO (dBm)		
		+4	+7	+10
10.1	40.1	20.58	19.55	21.69
49.8	79.8	17.10	23.21	24.61
89.5	119.5	25.98	23.46	19.70
129.2	159.2	15.70	17.45	18.71
168.9	198.9	20.64	19.21	22.06
208.6	238.6	16.94	18.28	19.60
248.3	278.3	19.23	23.39	21.01
287.9	317.9	14.94	19.13	19.37
327.6	357.6	16.18	19.14	16.68
367.3	397.3	16.07	18.93	18.18
407.0	437.0	20.34	16.63	17.56
446.7	476.7	25.48	19.66	21.97
486.4	516.4	21.90	17.04	14.66
526.1	556.1	20.86	18.40	19.56
565.8	595.8	16.55	25.28	18.45
585.6	615.6	18.02	15.13	17.39
625.3	655.3	16.12	23.01	16.59
645.2	675.2	29.42	22.38	19.77
684.9	714.9	17.40	16.77	14.88
704.7	734.7	19.49	15.25	16.50
744.4	774.4	18.66	20.32	21.44
764.3	794.3	18.26	16.03	20.65
803.9	833.9	18.84	19.35	23.37
823.8	853.8	15.75	16.30	17.23
863.5	893.5	15.36	21.97	15.81
883.3	913.3	12.82	18.42	15.18
923.0	953.0	11.48	17.00	17.95
942.9	972.9	11.45	16.53	18.52
982.6	1012.6	12.25	16.90	15.29
1002.4	1032.4	11.37	15.07	22.53
1042.1	1072.1	12.72	16.82	29.44
1061.9	1091.9	10.88	16.74	21.45
1101.6	1131.6	11.45	18.80	17.17
1121.5	1151.5	11.48	16.69	20.85
1161.2	1191.2	12.29	17.70	14.82
1181.0	1211.0	13.23	16.90	17.88
1220.7	1250.7	13.35	17.19	15.98
1240.6	1270.6	12.98	13.03	16.18
1280.3	1310.3	12.57	12.64	14.42
1300.1	1330.1	11.71	12.45	14.13

RF (IN) (MHz)	LO (MHz)	COMPRESSION @RF IN=+1dBm (dB)		
		@LO (dBm)		
		+4	+7	+10
10.1	40.1	0.55	0.41	0.33
49.8	79.8	0.53	0.43	0.26
89.5	119.5	0.50	0.34	0.33
129.2	159.2	0.55	0.38	0.24
168.9	198.9	0.57	0.41	0.19
208.6	238.6	0.48	0.35	0.31
248.3	278.3	0.58	0.42	0.28
287.9	317.9	0.59	0.38	0.28
327.6	357.6	0.61	0.46	0.26
367.3	397.3	0.63	0.49	0.33
407.0	437.0	0.66	0.49	0.25
446.7	476.7	0.63	0.46	0.30
486.4	516.4	0.65	0.46	0.29
526.1	556.1	0.65	0.41	0.30
565.8	595.8	0.65	0.40	0.27
585.6	615.6	0.60	0.45	0.35
625.3	655.3	0.59	0.38	0.23
645.2	675.2	0.82	0.63	0.45
684.9	714.9	0.79	0.43	0.46
704.7	734.7	0.76	0.50	0.40
744.4	774.4	0.91	0.57	0.43
764.3	794.3	0.82	0.58	0.40
803.9	833.9	0.87	0.55	0.50
823.8	853.8	1.04	0.80	0.59
863.5	893.5	0.88	0.76	0.52
883.3	913.3	0.86	0.64	0.53
923.0	953.0	1.06	0.74	0.58
942.9	972.9	1.17	0.92	0.72
982.6	1012.6	1.09	0.98	0.76
1002.4	1032.4	1.22	0.90	0.74
1042.1	1072.1	1.14	0.85	0.74
1061.9	1091.9	1.02	0.88	0.80
1101.6	1131.6	1.04	0.85	0.73
1121.5	1151.5	0.99	0.82	0.76
1161.2	1191.2	1.07	0.87	0.74
1181.0	1211.0	1.09	0.91	0.73
1220.7	1250.7	1.09	0.82	0.57
1240.6	1270.6	1.05	0.84	0.70
1280.3	1310.3	0.94	0.77	0.68
1300.1	1330.1	1.17	1.00	0.73

REV. X2

JMS-2+

100817

Page 1 of 5



IF/RF MICROWAVE COMPONENTS • ISO 9001 ISO 14001 AS9100 CERTIFIED • RoHS compliant

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Typical Performance Data

IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=500.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=20.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=1000.1MHz (dB)
		@LO (dBm)			@LO (dBm)			@LO (dBm)
		+7			+7			+7
460.0	40.1	7.03	10.0	30.1	7.38	980.0	20.1	8.00
448.8	51.4	6.99	29.4	49.5	6.84	960.6	39.5	7.92
437.5	62.6	7.02	48.8	68.9	6.86	941.2	58.9	7.87
426.3	73.9	7.00	68.2	88.3	6.87	921.8	78.3	7.84
415.0	85.1	6.96	87.6	107.7	6.89	902.4	97.7	7.78
403.8	96.4	6.90	107.0	127.1	6.89	883.0	117.1	7.72
392.5	107.6	6.88	126.4	146.5	6.90	863.6	136.5	7.66
381.3	118.9	6.86	145.8	165.9	6.93	844.2	155.9	7.67
370.0	130.1	6.81	165.2	185.3	6.94	824.8	175.3	7.63
358.8	141.4	6.80	184.6	204.7	6.96	805.4	194.7	7.61
347.5	152.6	6.81	204.0	224.1	6.95	786.0	214.1	7.59
336.3	163.9	6.79	223.4	243.5	6.98	766.6	233.5	7.55
325.0	175.1	6.78	242.8	262.9	7.00	747.2	252.9	7.56
313.8	186.4	6.78	262.2	282.3	7.03	727.8	272.3	7.55
302.5	197.6	6.78	281.6	301.7	7.05	708.4	291.7	7.53
291.3	208.9	6.77	301.0	321.1	7.05	689.0	311.1	7.48
280.0	220.1	6.76	320.4	340.5	7.07	669.6	330.5	7.45
268.8	231.4	6.75	339.8	359.9	7.07	650.2	349.9	7.47
257.5	242.6	6.77	378.6	398.7	7.18	611.4	388.7	7.47
246.3	253.9	6.75	398.0	418.1	7.13	592.0	408.1	7.43
235.0	265.1	6.71	436.8	456.9	7.39	553.2	446.9	7.43
223.8	276.4	6.71	456.2	476.3	7.23	533.8	466.3	7.42
212.5	287.6	6.79	495.0	515.1	7.23	495.0	505.1	7.45
201.3	298.9	6.78	514.4	534.5	7.24	475.6	524.5	7.44
190.0	310.1	6.80	553.2	573.3	7.33	436.8	563.3	7.50
178.8	321.4	6.78	572.6	592.7	7.32	417.4	582.7	7.50
167.5	332.6	6.79	611.4	631.5	7.41	378.6	621.5	7.40
156.3	343.9	6.78	630.8	650.9	7.41	359.2	640.9	7.41
145.0	355.1	6.81	669.6	689.7	7.43	320.4	679.7	7.45
133.8	366.4	6.83	689.0	709.1	7.44	301.0	699.1	7.41
122.5	377.6	6.83	727.8	747.9	7.48	262.2	737.9	7.43
111.3	388.9	6.85	747.2	767.3	7.47	242.8	757.3	7.45
100.0	400.1	6.83	786.0	806.1	7.49	204.0	796.1	7.46
88.8	411.4	6.84	805.4	825.5	7.47	184.6	815.5	7.45
77.5	422.6	6.82	844.2	864.3	7.51	145.8	854.3	7.51
66.3	433.9	6.88	863.6	883.7	7.47	126.4	873.7	7.50
55.0	445.1	6.89	902.4	922.5	7.48	87.6	912.5	7.53
43.8	456.4	6.87	921.8	941.9	7.48	68.2	931.9	7.54
21.3	478.9	6.95	960.6	980.7	7.48	29.4	970.7	7.60
10.0	490.1	7.36	980.0	1000.1	7.45	10.0	990.1	7.67

Typical Performance Data

LO (MHz)	LO-RF ISOLATION (dB)			LO-IF ISOLATION (dB)		
	@LO (dBm)			@LO (dBm)		
	+4	+7	+10	+4	+7	+10
10.1	60.50	68.19	68.66	62.58	65.18	62.34
49.8	59.41	57.79	58.18	53.32	51.93	51.37
89.5	54.55	54.38	54.14	48.98	47.36	46.61
129.2	51.89	52.18	51.88	45.77	44.68	43.87
168.9	50.48	50.77	50.45	43.96	42.86	42.09
208.6	49.59	49.68	49.11	42.57	41.55	40.79
248.3	48.82	48.76	48.14	41.52	40.50	39.63
287.9	48.33	47.94	47.20	40.69	39.68	38.78
327.6	47.72	47.18	46.36	39.96	38.87	37.94
367.3	46.95	46.30	45.48	39.15	38.07	37.19
407.0	46.04	45.40	44.64	38.60	37.60	36.75
446.7	45.06	44.51	43.86	37.85	36.89	36.05
486.4	43.92	43.47	42.94	37.31	36.47	35.75
526.1	42.95	42.68	42.26	36.69	35.79	35.15
565.8	41.82	41.70	41.46	36.25	35.33	34.55
585.6	41.32	41.25	41.09	36.11	35.28	34.44
625.3	40.16	40.22	40.16	35.32	34.85	34.20
645.2	39.66	39.80	39.78	34.99	34.63	34.14
684.9	38.70	38.96	39.05	34.33	34.02	33.78
704.7	38.13	38.47	38.62	33.90	33.55	33.35
744.4	37.21	37.67	37.95	33.17	32.66	32.42
764.3	36.77	37.28	37.62	33.00	32.39	32.05
803.9	35.89	36.48	36.91	32.63	31.96	31.37
823.8	35.54	36.18	36.64	32.47	31.89	31.23
863.5	34.86	35.55	36.10	32.15	32.06	31.44
883.3	34.54	35.25	35.82	31.93	32.08	31.55
923.0	34.04	34.83	35.46	31.22	31.77	31.54
942.9	33.74	34.54	35.19	30.98	31.75	31.78
982.6	33.19	34.05	34.75	30.40	31.36	31.74
1002.4	32.90	33.79	34.54	30.18	31.20	31.70
1042.1	32.36	33.28	34.06	29.61	30.70	31.44
1061.9	32.16	33.10	33.89	29.63	30.72	31.50
1101.6	31.63	32.61	33.42	29.10	30.22	31.17
1121.5	31.38	32.37	33.19	28.97	30.03	31.00
1161.2	30.86	31.91	32.75	28.62	29.60	30.66
1181.0	30.55	31.64	32.50	28.45	29.29	30.30
1220.7	29.95	31.08	31.99	28.30	28.90	29.72
1240.6	29.76	30.91	31.85	28.46	28.97	29.71
1280.3	29.09	30.26	31.24	28.87	29.18	29.66
1300.1	28.83	30.00	31.01	29.21	29.49	29.87

RF (IN) (MHz)	LO (MHz)	RF-IF ISOLATION (dB)		
		@LO (dBm)		
		+4	+7	+10
10.1	40.1	36.05	35.83	40.87
49.8	79.8	36.54	35.78	36.51
89.5	119.5	31.32	31.19	30.60
129.2	159.2	28.27	28.19	28.28
168.9	198.9	26.37	26.33	26.26
208.6	238.6	24.97	24.89	24.83
248.3	278.3	23.95	23.86	23.84
287.9	317.9	23.18	23.14	23.11
327.6	357.6	22.56	22.54	22.52
367.3	397.3	22.06	22.07	22.06
407.0	437.0	21.72	21.79	21.80
446.7	476.7	21.40	21.49	21.54
486.4	516.4	21.06	21.11	21.20
526.1	556.1	20.80	20.83	20.86
565.8	595.8	20.43	20.47	20.48
585.6	615.6	20.23	20.30	20.33
625.3	655.3	19.68	19.80	19.88
645.2	675.2	19.36	19.49	19.59
684.9	714.9	18.64	18.76	18.88
704.7	734.7	18.28	18.42	18.52
744.4	774.4	17.48	17.62	17.71
764.3	794.3	17.14	17.26	17.34
803.9	833.9	16.39	16.47	16.53
823.8	853.8	16.08	16.16	16.22
863.5	893.5	15.45	15.51	15.57
883.3	913.3	15.16	15.22	15.28
923.0	953.0	14.68	14.72	14.78
942.9	972.9	14.45	14.50	14.55
982.6	1012.6	14.02	14.06	14.13
1002.4	1032.4	13.86	13.90	13.96
1042.1	1072.1	13.53	13.58	13.66
1061.9	1091.9	13.42	13.47	13.55
1101.6	1131.6	13.14	13.21	13.28
1121.5	1151.5	13.04	13.11	13.19
1161.2	1191.2	12.78	12.86	12.93
1181.0	1211.0	12.70	12.78	12.84
1220.7	1250.7	12.56	12.65	12.69
1240.6	1270.6	12.48	12.59	12.66
1280.3	1310.3	12.41	12.55	12.65
1300.1	1330.1	12.38	12.54	12.66

REV. X2

JMS-2+

100817

Page 3 of 5



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Typical Performance Data

RF (IN) (MHz)	LO (MHz)	RF VSWR (:1)			LO (MHz)	LO VSWR (:1)			IF (OUT) (MHz)	IF VSWR @LO=1001.1MHz (:1)		
		@LO (dBm)				@LO (dBm)				@LO (dBm)		
		+4	+7	+10		+4	+7	+10		+4	+7	+10
10.1	40.1	1.33	1.22	1.16	10.1	1.61	2.22	3.12	10.1	3.77	1.96	2.03
49.8	79.8	1.32	1.39	1.44	49.8	1.84	2.63	3.65	30.3	2.00	1.75	1.83
89.5	119.5	1.17	1.24	1.30	89.5	1.83	2.60	3.60	50.5	1.85	1.66	1.45
129.2	159.2	1.22	1.32	1.38	129.2	1.80	2.55	3.50	70.7	1.87	1.58	1.41
168.9	198.9	1.16	1.24	1.30	168.9	1.84	2.59	3.55	90.9	1.99	1.71	1.53
208.6	238.6	1.25	1.34	1.39	208.6	1.79	2.51	3.41	111.1	2.08	1.77	1.60
248.3	278.3	1.24	1.33	1.40	248.3	1.84	2.57	3.50	131.3	2.02	1.72	1.57
287.9	317.9	1.24	1.33	1.39	287.9	1.84	2.56	3.48	151.5	1.95	1.66	1.52
327.6	357.6	1.29	1.37	1.43	327.6	1.84	2.58	3.49	171.7	1.93	1.68	1.53
367.3	397.3	1.28	1.37	1.43	367.3	1.90	2.66	3.60	191.9	1.99	1.74	1.59
407.0	437.0	1.31	1.39	1.45	407.0	1.86	2.58	3.47	212.1	1.99	1.73	1.58
446.7	476.7	1.30	1.38	1.44	446.7	1.90	2.63	3.54	232.3	1.96	1.71	1.56
486.4	516.4	1.31	1.39	1.44	486.4	1.88	2.58	3.45	252.5	1.92	1.68	1.55
526.1	556.1	1.31	1.38	1.44	526.1	1.90	2.60	3.47	272.8	1.92	1.70	1.57
565.8	595.8	1.35	1.43	1.49	565.8	1.94	2.65	3.52	293.0	1.96	1.73	1.58
585.6	615.6	1.34	1.42	1.48	585.6	1.92	2.61	3.47	313.2	1.91	1.69	1.55
625.3	655.3	1.42	1.51	1.58	625.3	1.96	2.67	3.52	333.4	1.87	1.66	1.52
645.2	675.2	1.47	1.56	1.63	645.2	2.00	2.73	3.60	353.6	1.84	1.65	1.54
684.9	714.9	1.47	1.55	1.62	684.9	1.99	2.69	3.52	373.8	1.87	1.66	1.56
704.7	734.7	1.54	1.63	1.70	704.7	2.00	2.68	3.51	394.0	1.89	1.69	1.59
744.4	774.4	1.62	1.71	1.78	744.4	2.07	2.77	3.61	434.4	1.83	1.65	1.57
764.3	794.3	1.59	1.68	1.74	764.3	2.03	2.70	3.50	454.6	1.83	1.66	1.57
803.9	833.9	1.66	1.75	1.82	803.9	2.05	2.71	3.49	495.0	1.85	1.69	1.61
823.8	853.8	1.68	1.76	1.83	823.8	2.11	2.80	3.61	515.2	1.86	1.68	1.61
863.5	893.5	1.57	1.63	1.69	863.5	2.12	2.77	3.56	555.6	1.83	1.69	1.63
883.3	913.3	1.59	1.66	1.73	883.3	2.13	2.78	3.56	575.8	1.87	1.73	1.67
923.0	953.0	1.63	1.69	1.74	923.0	2.21	2.89	3.69	616.2	1.84	1.71	1.66
942.9	972.9	1.57	1.62	1.67	942.9	2.21	2.86	3.64	636.4	1.83	1.71	1.68
982.6	1012.6	1.64	1.70	1.76	982.6	2.20	2.82	3.56	676.8	1.86	1.74	1.70
1002.4	1032.4	1.71	1.76	1.82	1002.4	2.23	2.86	3.62	697.0	1.85	1.73	1.70
1042.1	1072.1	1.71	1.76	1.81	1042.1	2.29	2.92	3.68	737.4	1.84	1.74	1.71
1061.9	1091.9	1.80	1.84	1.90	1061.9	2.32	2.96	3.72	757.7	1.86	1.76	1.73
1101.6	1131.6	2.00	2.04	2.08	1101.6	2.31	2.94	3.67	798.1	1.83	1.72	1.70
1121.5	1151.5	2.04	2.06	2.10	1121.5	2.35	2.97	3.70	818.3	1.82	1.72	1.70
1161.2	1191.2	2.27	2.29	2.32	1161.2	2.39	3.00	3.73	858.7	1.86	1.75	1.74
1181.0	1211.0	2.39	2.39	2.41	1181.0	2.34	2.94	3.64	878.9	1.84	1.74	1.71
1220.7	1250.7	2.44	2.43	2.44	1220.7	2.38	2.97	3.68	919.3	1.80	1.71	1.69
1240.6	1270.6	2.55	2.53	2.53	1240.6	2.45	3.06	3.79	939.5	1.83	1.72	1.71
1280.3	1310.3	2.62	2.59	2.57	1280.3	2.44	3.01	3.70	979.9	1.83	1.71	1.68
1300.1	1330.1	2.58	2.56	2.54	1300.1	2.45	3.01	3.70	1000.1	1.87	1.83	1.87

REV. X2

JMS-2+

100817

Page 4 of 5



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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	7	28	16	24	26	33	37	32	36	50
1	-	14	0	30	12	38	19	38	37	41	50	37
2	110	61	52	70	53	64	52	71	57	77	69	60
3	112	83	75	85	73	78	64	87	77	83	75	79
4	110	98	95	96	90	83	101	101	94	104	92	99
5	121	108	104	101	102	91	100	105	104	98	103	107
6	113	103	104	106	104	100	93	88	97	105	104	98
7	119	107	114	122	110	104	106	101	93	103	105	106
8	117	113	99	102	110	99	97	108	91	92	97	99
9	119	103	104	102	104	98	108	105	96	92	87	101
10	119	101	107	110	111	107	105	105	113	93	99	91
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions: RF IN: 500.1 MHz; -14.00 dBm.
LO IN: 530.01 MHz; +7.00 dBm
IF OUT: 29.91 MHz; -20.87 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	17	38	27	34	40	44	50	47	52	76
1	-	14	0	30	12	39	20	41	37	46	57	43
2	97	51	44	56	43	54	45	60	51	64	67	56
3	119	61	51	62	54	55	46	60	49	57	58	57
4	112	72	78	75	66	81	68	81	59	74	63	78
5	114	77	76	76	63	75	59	71	58	77	60	80
6	109	103	85	92	81	85	78	83	74	84	75	89
7	127	98	89	98	79	93	79	90	79	88	84	83
8	115	103	108	108	106	103	101	118	91	95	93	97
9	117	102	110	108	104	102	103	106	101	102	91	115
10	118	110	117	113	114	111	103	105	116	110	105	105
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions: RF IN: 500.1 MHz; -4.00 dBm.
LO IN: 530.01 MHz; +7.00 dBm
IF OUT: 29.91 MHz; -11.05 dBm

- Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.
2. + entry denotes harmonics are in (dBc) above IF OUTPUT.
3. RF Cal represent the Harmonics level of the RF input signal to the mixer.



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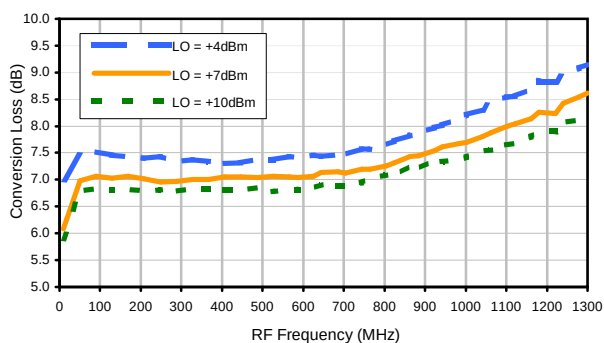
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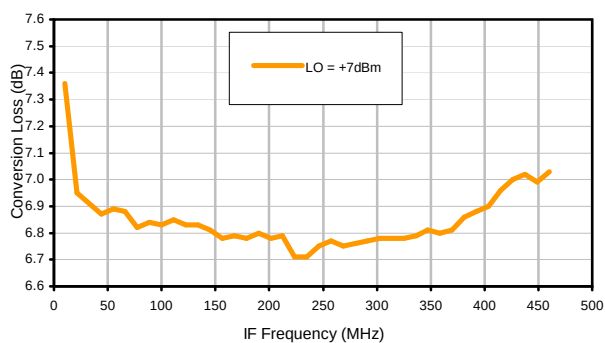
REV. X2
JMS-2+
100817
Page 5 of 5

Typical Performance Curves

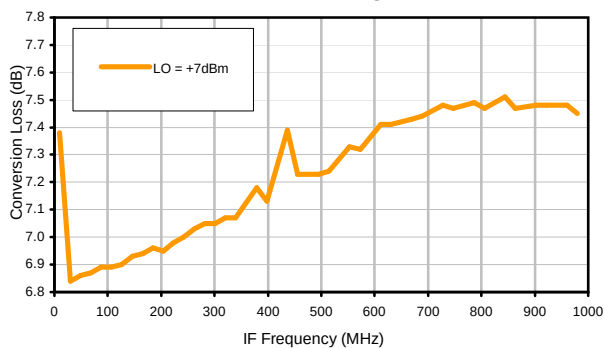
Conversion Loss @ IF=30MHz



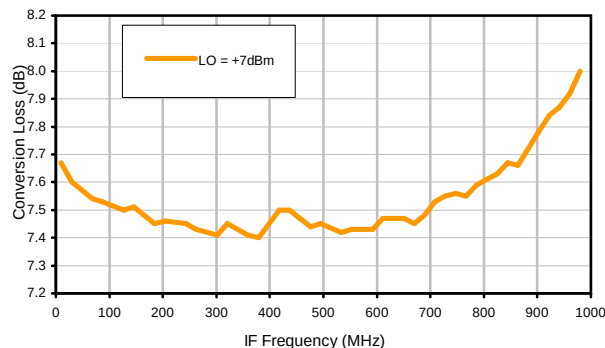
Conversion Loss vs. IF @ RF=500.1MHz



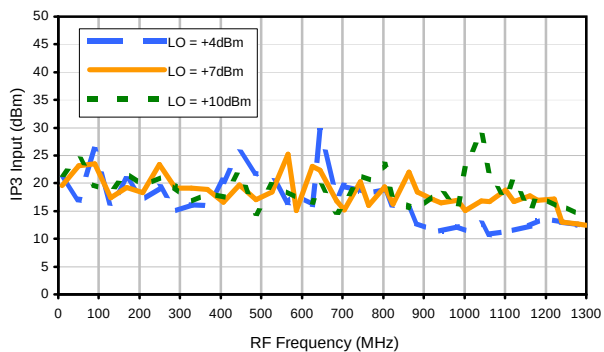
Conversion Loss vs. IF @ RF=20.1MHz



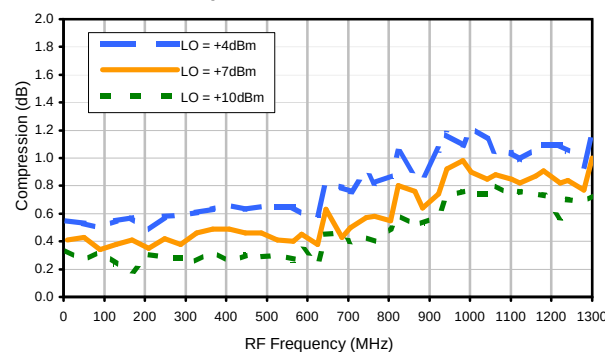
Conversion Loss vs. IF @ RF=1000.1MHz



IP3 Input

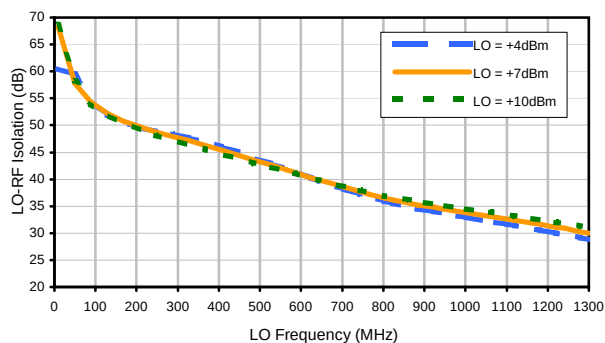


Compression @ RF IN=+1dBm

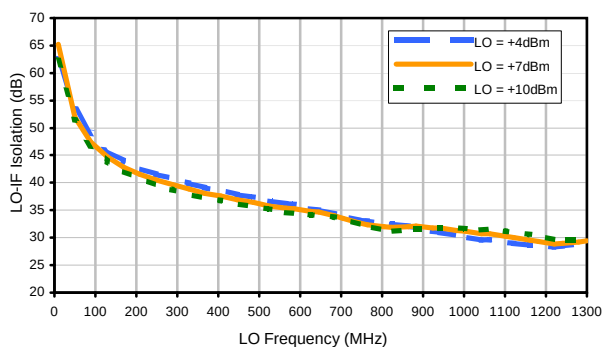


Typical Performance Curves

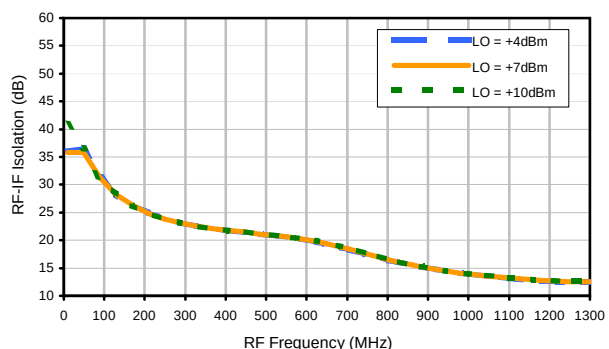
LO-RF Isolation



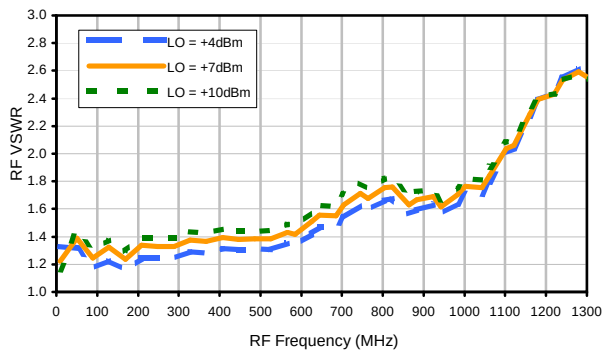
LO-IF Isolation



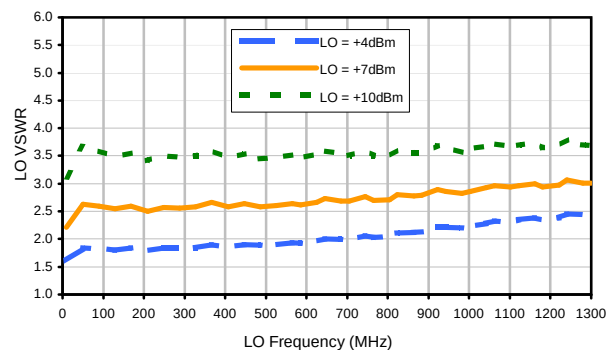
RF-IF Isolation



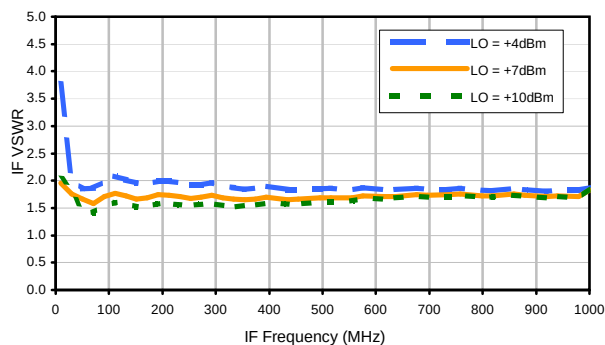
RF VSWR



LO VSWR



IF VSWR



REV. X2

JMS-2+

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Page 2 of 3



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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	7	28	16	24	26	33	37	32	36	50
1	-	14	0	30	12	38	19	38	37	41	50	37
2	110	61	52	70	53	64	52	71	57	77	69	60
3	112	83	75	85	73	78	64	87	77	83	75	79
4	110	98	95	96	90	83	101	101	94	104	92	99
5	121	108	104	101	102	91	100	105	104	98	103	107
6	113	103	104	106	104	100	93	88	97	105	104	98
7	119	107	114	122	110	104	106	101	93	103	105	106
8	117	113	99	102	110	99	97	108	91	92	97	99
9	119	103	104	102	104	98	108	105	96	92	87	101
10	119	101	107	110	111	107	105	105	113	93	99	91
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 500.1 MHz; -14.00 dBm.

LO IN: 530.01 MHz; +7.00 dBm

IF OUT: 29.91 MHz; -20.87 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	17	38	27	34	40	44	50	47	52	76
1	-	14	0	30	12	39	20	41	37	46	57	43
2	97	51	44	56	43	54	45	60	51	64	67	56
3	119	61	51	62	54	55	46	60	49	57	58	57
4	112	72	78	75	66	81	68	81	59	74	63	78
5	114	77	76	76	63	75	59	71	58	77	60	80
6	109	103	85	92	81	85	78	83	74	84	75	89
7	127	98	89	98	79	93	79	90	79	88	84	83
8	115	103	108	108	106	103	101	118	91	95	93	97
9	117	102	110	108	104	102	103	106	101	102	91	115
10	118	110	117	113	114	111	103	105	116	110	105	105
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 500.1 MHz; -4.00 dBm.

LO IN: 530.01 MHz; +7.00 dBm

IF OUT: 29.91 MHz; -11.05 dBm

Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.

2. + entry denotes harmonics are in (dBc) above IF OUTPUT.

3. RF Cal represent the Harmonics level of the RF input signal to the mixer.

REV. X2

JMS-2+

100817

Page 3 of 3



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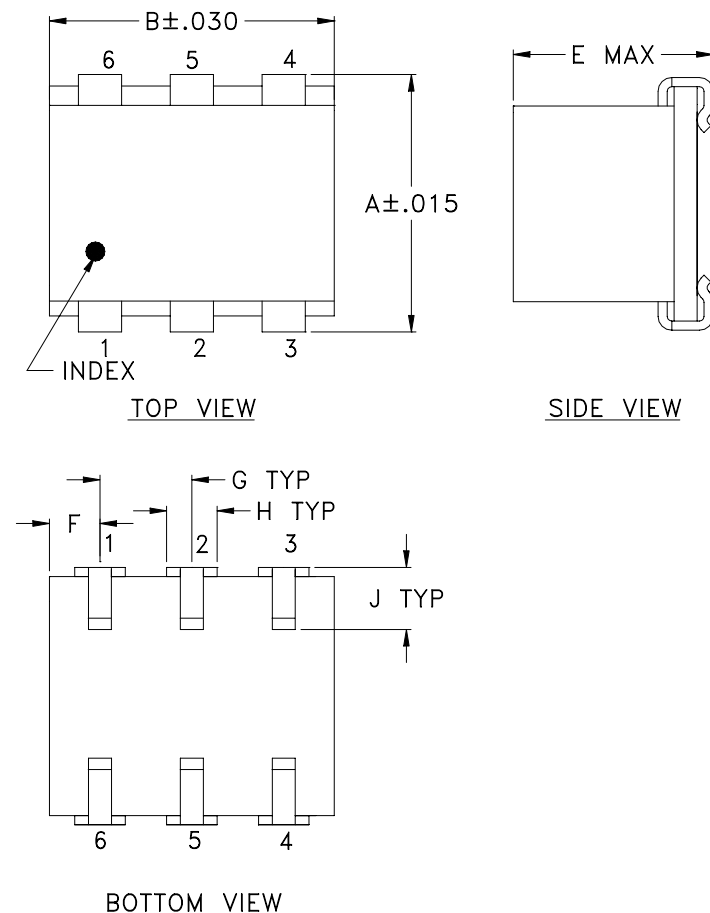
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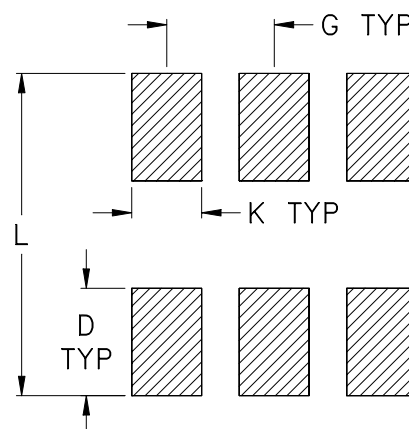
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Outline Dimensions



PCB Land Pattern



Suggested Layout,
Tolerance to be within $\pm .002$

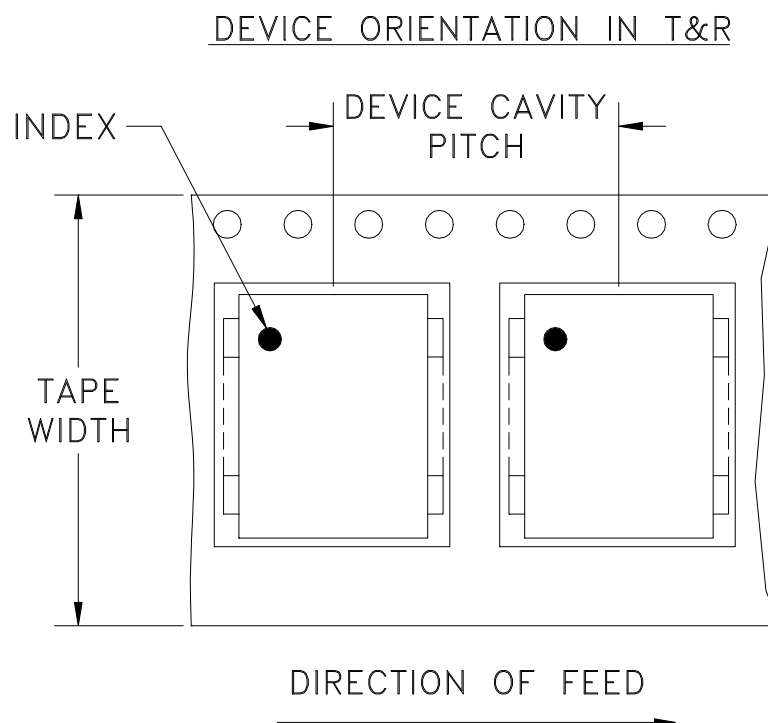
CASE #	A	B	C	D	E	F	G	H	J	K	L	WT. GRAM
BH292	.280 (7.11)	.310 (7.87)	- -	.100 (2.54)	.225 (5.72)	.055 (1.40)	.100 (2.54)	.047 (1.19)	.065 (1.65)	.065 (1.65)	.300 (7.62)	.45

Dimensions are in inches (mm). Tolerances: 2Pl. $\pm .01$; 3Pl. $\pm .005$

Notes:

- Case material: Ceramic.
- Termination finish:
For RoHS Case Styles: Tin plate over Nickel plate. All models, (+) suffix.
For RoHS-5 Case Styles: Tin-Lead plate. All models, no (+) suffix.

Tape & Reel Packaging TR-F24



Tape Width, mm	Device Cavity Pitch, mm	Reel Size, inches	Devices per Reel
16	12	13	500

Mini-Circuits carrier tape materials provide protection from ESD (Electro-Static Discharge) during handling and transportation. Tapes are static dissipative and comply with industry standards EIA-481/EIA-541.

Go to: www.minicircuits.com/pages/pdfs/tape.pdf



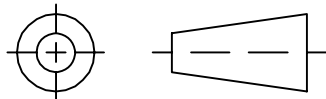
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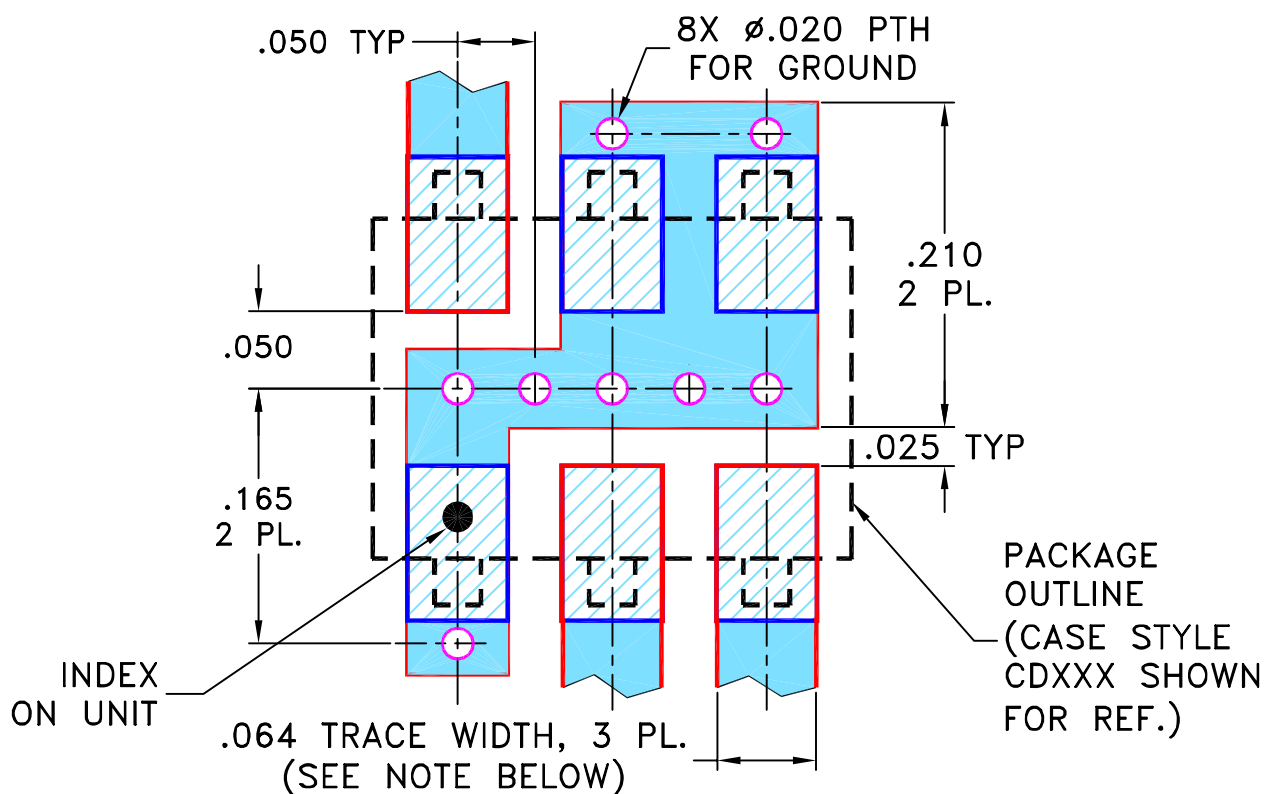
THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
A	M101143	ADDED "gk" PIN CONNECTION, TT100 CASE STYLE & NOTE 2	10/10/05	MMG	DJ
B	M102713	ADDED "...WITH SMOBC"	01/17/06	MMG	IL
C	M108637	REMOVED "PIN 1", ADDED INDEX ON UNIT	12/01/06	MYG	FL

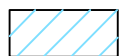
SUGGESTED MOUNTING CONFIGURATION
FOR BH292, CD541/542/636/637, TT100/240 CASE
STYLES, "gk", "ht", "hu", "nd", "w" PIN CONNECTIONS



- NOTES:** 1. TRACE WIDTH IS SHOWN FOR ROGERS R04350B WITH DIELECTRIC THICKNESS .030" $\pm$.002"; COPPER: 1/2 OZ. EACH SIDE.
 FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
 2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.



DENOTES PCB COPPER LAYOUT WITH SMOBC
 (SOLDER MASK OVER BARE COPPER)



DENOTES COPPER LAND PATTERN FREE OF SOLDER MASK

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

TOLERANCES ON:

2 PL DECIMALS $\pm$ 3 PL DECIMALS $\pm$.005ANGLES $\pm$ FRACTIONS $\pm$

DRAWN

MMG

07/17/02

CHECKED

WL

08/02/02

APPROVED

DJ

08/05/02



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 Brooklyn NY 11235

PL, gk/ht/hu/nd/w, BH292,
 CD541/542/636/637, TT100/240, TB-03

SIZE

A

CODE IDENT

15542

DRAWING NO:

98-PL-052

REV:

C

FILE:

98PL052

SCALE:

8:1

SHEET:

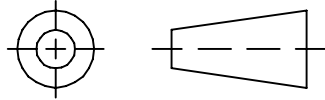
1 OF 1

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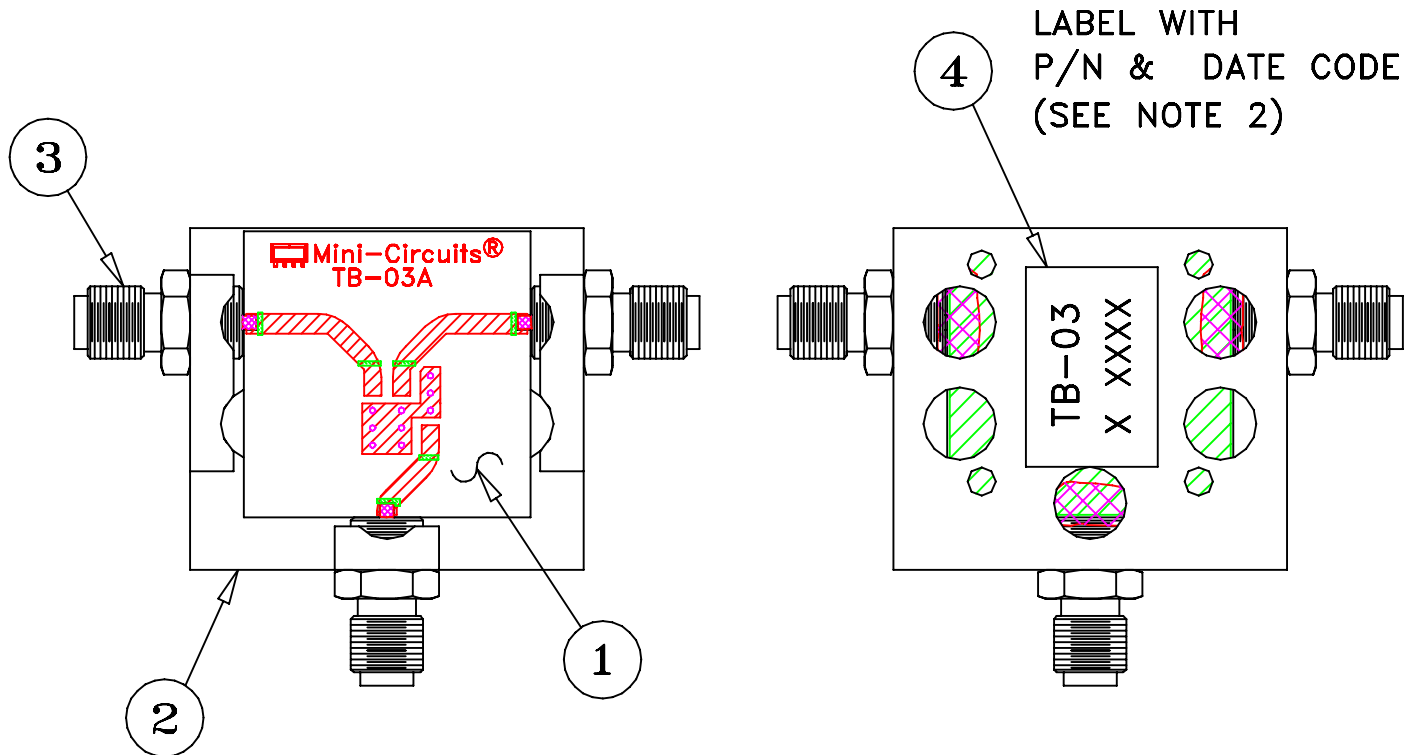
ASHEETA1.DWG REV:A DATE:01/12/95

THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
E	M119737	UPDATED PCB	10.08	MF	AD
F	M127659	UPDATED CARR	06.10	SW	SG
G	M127846	UPDATED SCHEMATIC DIAGRAM	06.10	SW	SG
H	M131840	UPDATED DWG	05.11	MF	AD

**NOTES:**

1. REFER TO -09 PAGE FOR ITEM DESCRIPTIONS.
DESIGNATION NUMBERS ON -20 PAGE CORRESPOND TO THE NUMBERS ON -09 PAGE.
2. FOR TEXT HEIGHT & STYLE ON THE LABEL REFER TO: D3-G209.

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

DRAWN

S.WOLYNSKI

08.29.99

TOLERANCES ON:

CHECKED

SG

07.06.99

2 PL DECIMALS ±

APPROVED

MG

07.10.99

3 PL DECIMALS ±

ANGLES ±

FRACTIONS ±

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ASHEETA1.DWG REV:A DATE:01/12/95


Mini-Circuits® 13 Neptune Avenue
Brooklyn NY 11235

TB,ADE,CD542/636,06MX01,50

SIZE

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CODE IDENT

15542

DRAWING NO:

TB-03-20

REV:

H

FILE:

WTB-03

SCALE:

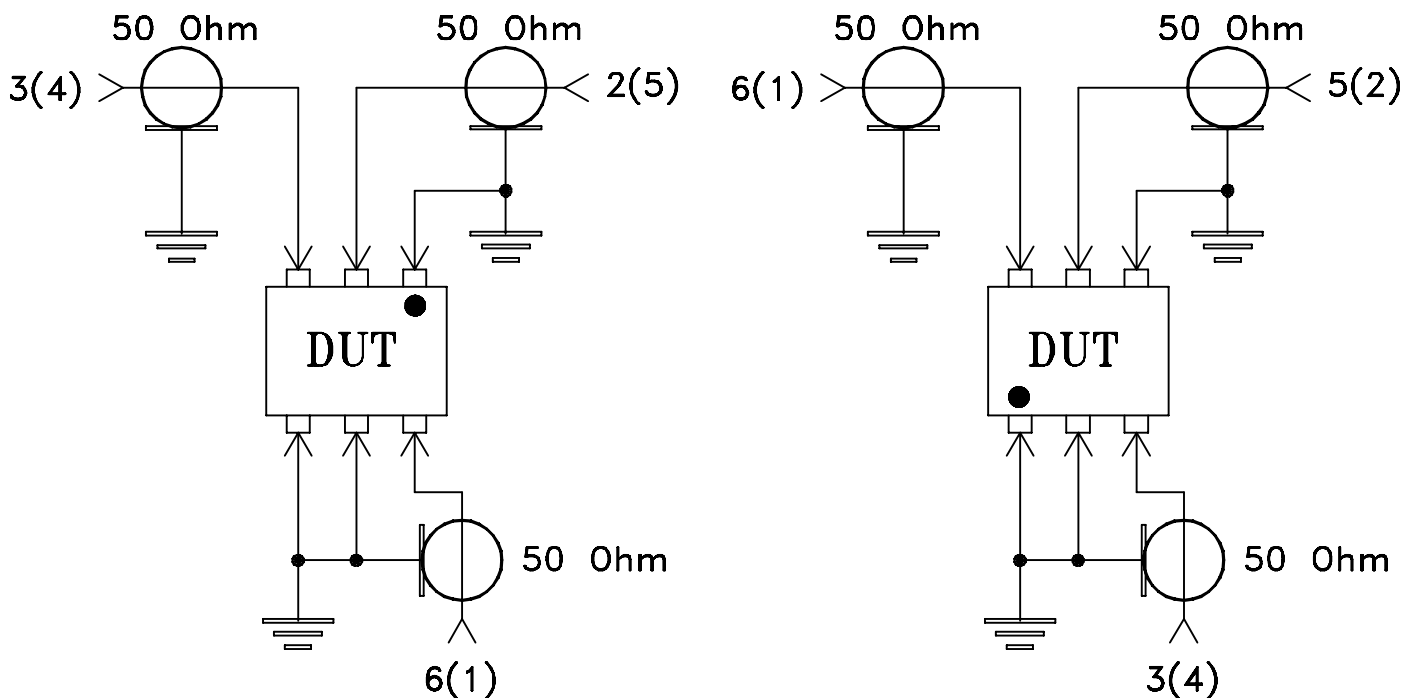
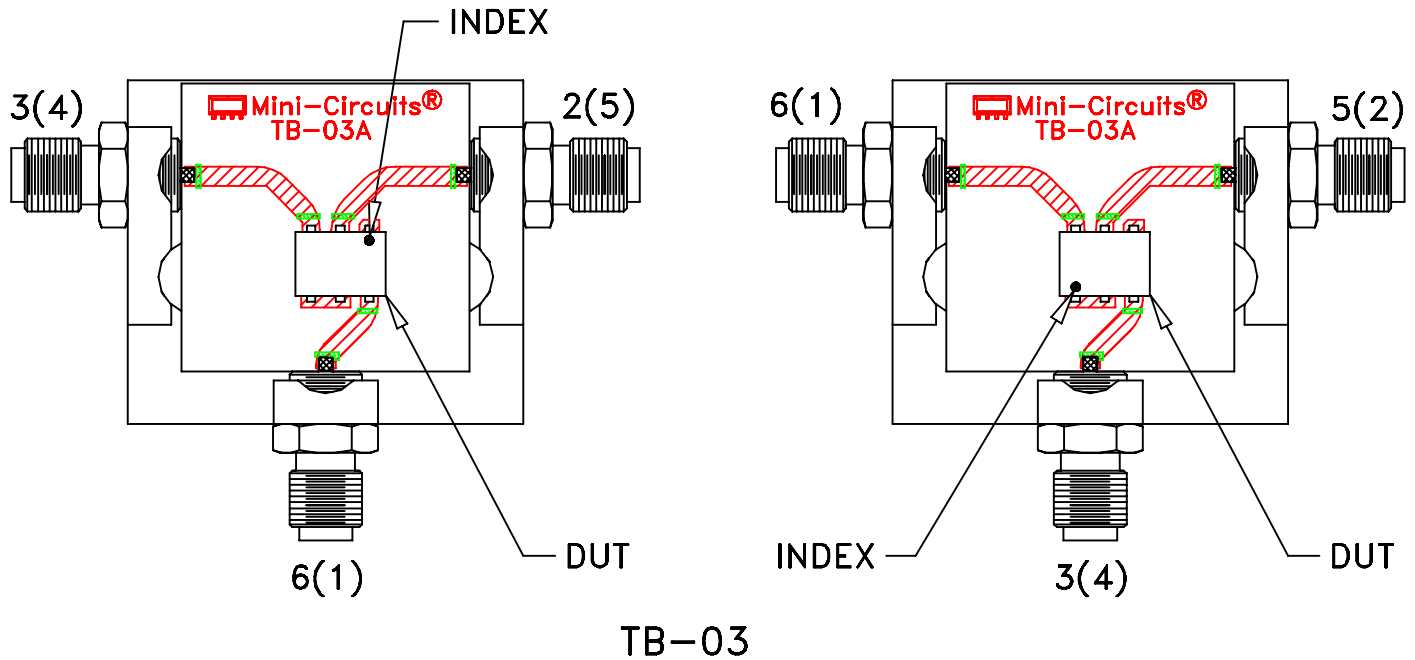
1.5:1

SHEET:

1 OF 2

Evaluation Board and Circuit

For Pin Connections and DUT Orientation Refer to
Data Sheet of the DUT



Schematic Diagram

Notes:

1. 50 Ohm SMA Female connectors.
2. PCB Material: Rogers R04350 or equivalent,
Dielectric Constant=3.5, Thickness=.030 inch.

 **Mini-Circuits®**



All Mini-Circuits products are manufactured under exacting quality assurance and control standards, and are capable of meeting published specifications after being subjected to any or all of the following physical and environmental test.

Specification	Test/Inspection Condition	Reference/Spec
Operating Temperature	-40° to 85°C Ambient Environment	Individual Model Data Sheet
Storage Temperature	-55° to 100° C Ambient Environment	Individual Model Data Sheet
Humidity	90 to 95% RH, 240 hours, 50°C	MIL-STD-202, Method 103, Condition A, Except 50°C and end-point electrical test done within 12 hours
Thermal Shock	-55° to 100°C, 100 cycles	MIL-STD-202, Method 107, Condition A-3, except +100°C
Solder Reflow Heat	Sn-Pb Eutetic Process: 225°C peak Pb-Free Process 245° - 250°C peak	J-STD-020, Table 4-1, 4-2 and 5-2, Figure 5-1
Solderability	10X Magnification	J-STD-002, 95% Coverage
Vibration (High Frequency)	20g peak, 10-2000 Hz, 12 times in each of three perpendicular directions (total 36)	MIL-STD-202, Method 204, Condition D
Mechanical Shock	50g, 11 ms, 1/2-sine, 18 shocks: 3 each direction, each of 3 axes	MIL-STD-202, Method 213, Condition A
Marking Resistance to Solvents	Isopropyl alcohol + mineral spirits at 25°C; terpene defluxer at 25°C; distilled water + proylene glycol monomethyl ether + monoethanolamine at 63°C to 70°C	MIL-STD-202, Method 215